# CSA1280 – COMPUTER ARCHITECTURE

## EXPERIMENT – 3

## ADDITION OF TWO 8-BIT NUMBERS WITH CARRY

**AIM:** To write and execute an 8085 assembly language program to subtract two 8-bit numbers and store the sum and carry in memory.

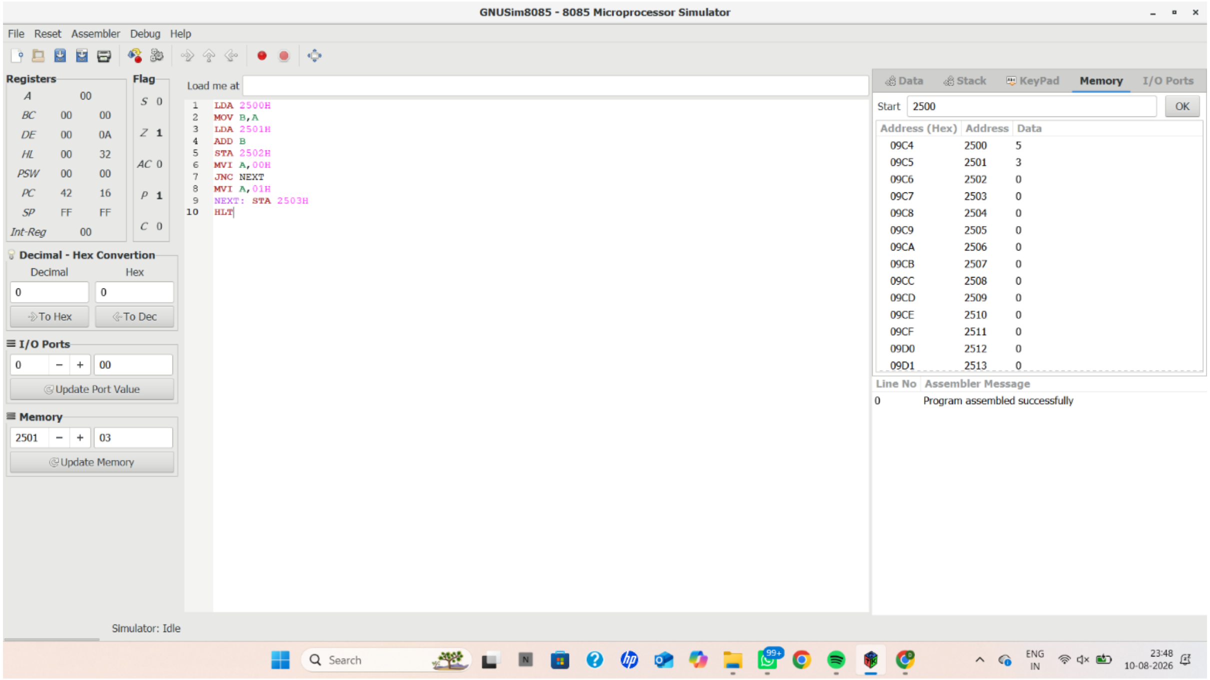
### ALGORITHM

1. Load the first number from 2500H.
2. Move it to register B.
3. Load the second number from 2501H.
4. Add register B with the accumulator.
5. Store the sum in 2502H.
6. Store the carry in 2503H.
7. Stop the program.

### CODE:

```
LDA 2500H
MOV B,A
LDA 2501H
ADD B
STA 2502H
MVI A,00H
JNC NEXT
MVI A,01H
NEXT: STA 2503H
HLT
```

# OUTPUT:


GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
A 00
BC 00 00
DE 00 0A
HL 00 32
PSW 00 00
PC 42 16
SP FF FF
Int-Reg 00
Flag
S 0
Z 1
AC 0
P 1
C 0
Decimal - Hex Convertion
Decimal Hex
To Hex To Dec
I/O Ports
Update Port Value
Memory
2501 03
Update Memory
Load me at
LDA 2500H
MOV B,A
LDA 2501H
ADD B
STA 2502H
MVI A,00H
JNC NEXT
MVI A,01H
NEXT: STA 2503H
HLT
Data Stack KeyPad Memory I/O Ports
Start 2500 OK
Address (Hex) Address Data
09C4 2500 5
09C5 2501 3
09C6 2502 0
09C7 2503 0
09C8 2504 0
09C9 2505 0
09CA 2506 0
09CB 2507 0
09CC 2508 0
09CD 2509 0
09CE 2510 0
09CF 2511 0
09D0 2512 0
09D1 2513 0
Line No Assembler Message
0 Program assembled successfully
Simulator: Idle
Search
ENG IN
23:48
10-08-2026